

Engineering Mathematics III

ENGE702

Lecture 3: Numerical methods and direction fields

Direction fields

For some (actually most) differential equations there is no (known) way of finding a nice expression for an exact solution. There are ways, however, that we can find approximate solutions and other general information about the solution.

The first method we will look at today is called a **direction field**. Sometimes these are called slope fields.

Recall that the derivative of a function is the slope of the function at a particular point. If a differential equation can be written in the form

$$y' = f(x, y),$$

we can plot the slopes of the solution function $y(x)$ for particular values of x and y .

This is best viewed in an example.

Direction fields

Example: Plot a direction field for the differential equation

$$y' + y = x + 1.$$

The question does not ask for a particular range of slopes, but we will plot for integer values of x and y between -3 and 3 .

The best way to start is to rewrite the equation in the form $y' = f(x, y)$, and then create a chart with the slopes for all the x and y values. From there it will be easy to plot.

Direction fields

For this example, we have $f'(x, y) = y' = x - y + 1$

$f'(x, y)$	$x = -3$	$x = -2$	$x = -1$	$x = 0$	$x = 1$	$x = 2$	$x = 3$
$y = 3$							
$y = 2$							
$y = 1$							
$y = 0$							
$y = -1$							
$y = -2$							
$y = -3$							

Now we can plot the slopes.

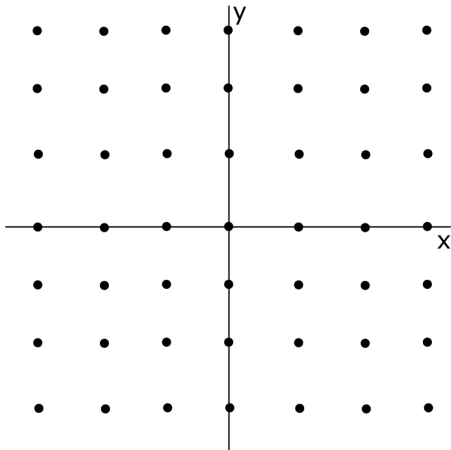
Direction fields

For this example, we have $y' = x - y + 1$

$f'(x, y)$	$x = -3$	$x = -2$	$x = -1$	$x = 0$	$x = 1$	$x = 2$	$x = 3$
$y = 3$	-5	-4	-3	-2	-1	0	1
$y = 2$	-4	-3	-2	-1	0	1	2
$y = 1$	-3	-2	-1	0	1	2	3
$y = 0$	-2	-1	0	1	2	3	4
$y = -1$	-1	0	1	2	3	4	5
$y = -2$	0	1	2	3	4	5	6
$y = -3$	1	2	3	4	5	6	7

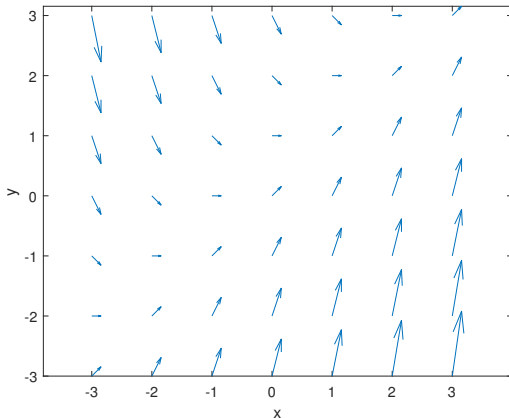
Now we can plot the slopes.

Direction fields



Direction fields

We get the following picture:



This helps us to visualise integrating forwards in x (or time t).

Direction fields

Direction fields can be plotted in MATLAB using quiver plots. For a function $y(x)$, an arrow at each point (x, y) in a subset of the Cartesian plane represents the gradient (or slope) y' .

If the horizontal component of the arrow is one, then the vertical component is y' .

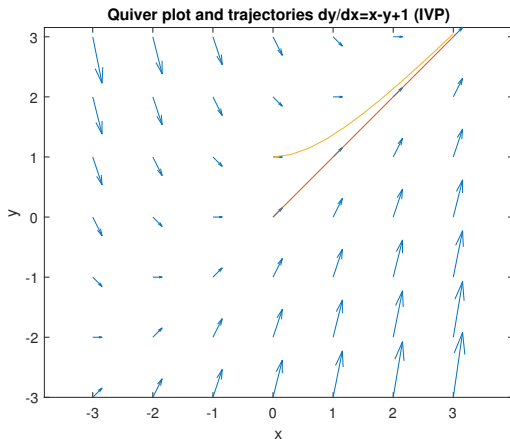
Direction fields

Now that we have some slopes plotted, we can plot some solutions for various initial conditions.

We will plot the solution curves for the initial conditions $y(0) = 0$ and $y(0) = 1$ on the sketch on the earlier slide.

Direction fields

We get the following picture:



The red and orange curves correspond to initial conditions $y(0) = 0$ and $y(0) = 1$, respectively.

The code

We create the quiver plot as follows. Practise with other starting points and other ODEs.

```
[X,Y] = meshgrid(-3:3,-3:3);
W=ones(7,7);
V=X-Y+W % vertical component of arrow,  $y'=x-y+1$ , 7 by 7 matrix
U=ones(7,7); % horizontal component of arrow
figure(1);
clf;
quiver(X,Y,U,V);
axis equal;

and plot a trajectory

x0=0;
xfinal=3;
y0=0;
[x,y]=ode23(@(x,y) (x-y+1),[x0 xfinal],y0);
figure(1)
hold on;
plot(x,y)
```

Euler's Method

The most basic numerical method for solving initial value problems is Euler's method. Again, we are assuming we can express our differential equation in the form $y' = f(x, y)$.

This method works as follows:

- ▶ We choose a step size. Generally a smaller step size will be more accurate, but take longer to compute.
- ▶ We start at our initial condition: a particular x value and y value.
- ▶ We calculate the slope at the initial point.
- ▶ We travel along the x direction a distance of the step size, and assume the solution function has the same slope for this duration.
- ▶ This will give us a new x value (the old one plus the step size), and a new y value (which is calculated assuming constant slope).
- ▶ We then calculate a new slope at this x - y value, and repeat the process as long as we need.

Euler's Method

Again, we will use an example to illustrate.

Example: Use Euler's method to find an approximate value for $y(4)$ for the initial value problem

$$y' + 2y = x, \quad y(0) = 1$$

We haven't been given a step size to use, but a step size of $h = 1$ will be used. We can re-write the equation as $y' = x - 2y$. Euler's method is often approached with a table as follows:

n	x_n	y_n	$f(x_n, y_n) = y'(x_n, y_n)$	$y_n + hf(x_n, y_n)$
0				
1				
2				
3				
4				

Note that $y_{n+1} = y_n + hf(x_n, y_n)$.

Euler's Method

Filling in the table:

$$y' = x - 2y, \quad y(0) = 1$$

A step size of $h = 1$ will be used.

n	x_n	y_n	$f(x_n, y_n)$	$y_n + hf(x_n, y_n)$
0	0	1	$x_0 - 2y_0 = -2$	-1
1	$x_1 = x_0 + h = 1$	-1	3	2
2	2	2	-2	0
3	3	0	3	3
4	4	3	-2	1

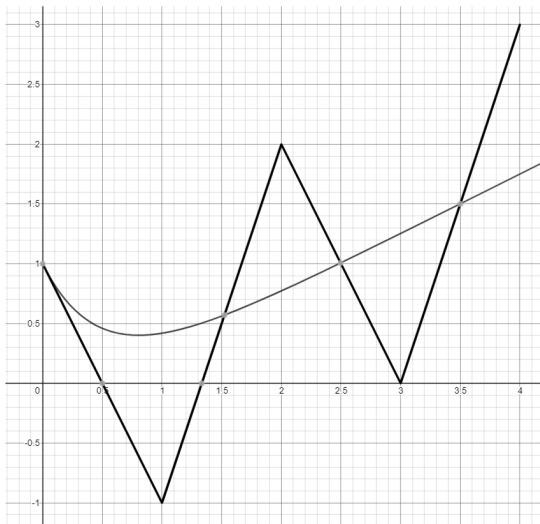
Note that $y_{n+1} = y_n + hf(x_n, y_n)$.

Euler's Method

The initial value problem we just saw has the solution

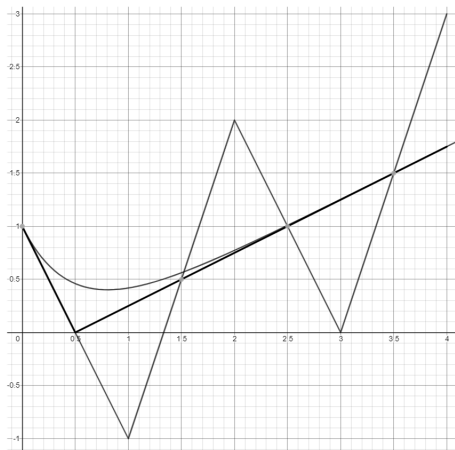
$$y = \frac{5}{4}e^{-2x} + \frac{x}{2} - \frac{1}{4} \text{ (we will see how to find this in a future lecture).}$$

Let's plot this against our estimates:



Euler's Method, smaller step size

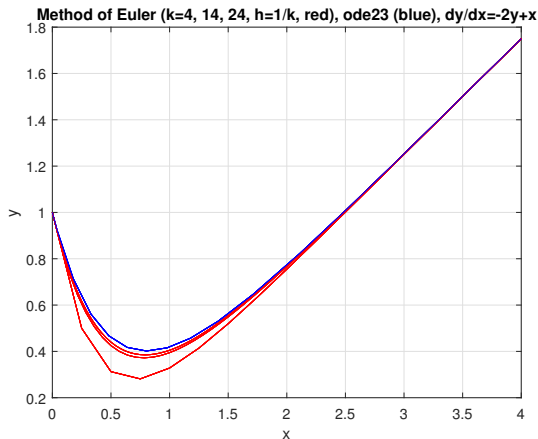
If we use a smaller step size what happens?



In this case we just happen to fall on a straight line solution to the general equation.

Euler's Method, even smaller step size

If we use an even smaller step size what happens?



As $h = 1/k$ decreases (i.e. as k increases) the accuracy improves (successive piecewise linear red plots better approximate the blue curve).

Exercises

Consider the differential equation

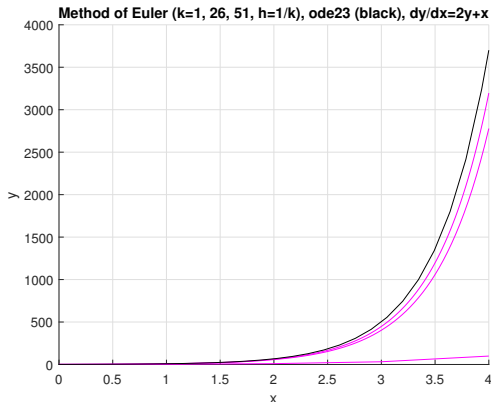
$$y' - 2y = x.$$

1. Plot a direction field for the differential equation
 - 1.1 by hand
 - 1.2 in MATLAB
2. Use Euler's method to estimate $y(4)$ given $y(0) = 1$. Use step sizes of:
 - 2.1 $h = 1$
 - 2.2 $h = 0.5$
 - 2.3 much smaller h

Exercises: Euler's method in MATLAB

Consider the same differential equation and much smaller step sizes

$$y' - 2y = x.$$



As $h = 1/k$ decreases (i.e. as k increases) the accuracy improves and the magenta piecewise linear plots (Euler's method) better approximate the black trajectory (ode23).

Text Book References

Chapter 1.2